DEPARTMENT OF ELECTRICAL ENGINEERING

COLLEGE OF E&ME, NUST, RAWALPINDI

Discrete Transistor Audio Amplifier using ±18V Supply

EE – 313 Electronic Circuit Design

PROJECT REPORT

DE – 46 – EE – A

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ABSTRACT

This project presents the design, simulation, and implementation of a discrete BJT-based audio amplifier capable of amplifying a low-level microphone signal to drive an 8 Ω loudspeaker. The amplifier operates on a ±18 V dual supply and achieves a voltage gain greater than 500 V/V with a bandwidth of 100 Hz to 12 kHz. The design uses only BJTs (no op-amps or IC audio amplifiers) and consists of an input buffer, two voltage gain stages, and a Class AB push-pull output stage. Simulation and hardware results confirm stable, low-distortion operation and compliance with design specifications.

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1. INTRODUCTION

Microphones produce millivolt-level signals that cannot directly drive speakers, so multi-stage amplification is required. Discrete transistor amplifiers are used in this project to understand biasing, gain distribution, and analog circuit behavior without relying on ICs.

Specifications:

- Supply: ±18 V
- Gain: ≥ 500 V/V
- Bandwidth: 100 Hz – 12 kHz
- Load: 8 Ω speaker
- Output Power: 1–3 W

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2. SYSTEM OVERVIEW

The signal path consists of:
Microphone → Input coupling & biasing → Emitter follower buffer → Two CE gain stages → Bias/level shifting network → Class AB output stage → Speaker

The signal is first buffered, then voltage amplified in two stages, and finally power-amplified to drive the load.

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3. POWER SUPPLY

A ±18 V dual supply is used for the output stage to allow symmetrical signal swing, eliminate large coupling capacitors, improve headroom, and reduce distortion. Small-signal stages use a +18 V single supply for simpler biasing.

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4. INPUT STAGE

The microphone signal is AC-coupled to block DC and biased using a resistor divider for stability. A capacitor is chosen to ensure a lower cutoff below 100 Hz. The input stage must maintain high input impedance to avoid signal loss.

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5. BUFFER STAGE (EMITTER FOLLOWER)

A common collector (emitter follower) is used as the first active stage. It provides high input impedance and low output impedance, ensuring proper signal transfer without loading the microphone.

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6. VOLTAGE AMPLIFICATION STAGES

Two cascaded common emitter stages provide the required voltage gain.

- First CE stage: moderate gain (~26 V/V) with stability via emitter degeneration
- Second CE stage: high gain (~25 V/V) to reach total gain requirement

Splitting gain across stages improves stability, reduces distortion, and maintains bandwidth.

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7. COUPLING & FREQUENCY RESPONSE

Coupling capacitors isolate DC between stages while passing AC signals. Component values are selected to maintain a flat response from 100 Hz to 12 kHz. Proper design ensures minimal signal loss at low frequencies.

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8. OUTPUT STAGE (CLASS AB PUSH-PULL)

A complementary Class AB push-pull stage delivers power to the 8 Ω speaker. It amplifies both positive and negative halves of the waveform efficiently while minimizing crossover distortion.

Emitter resistors improve thermal stability and current sharing. The dual supply enables full signal swing around ground and eliminates the need for an output capacitor.

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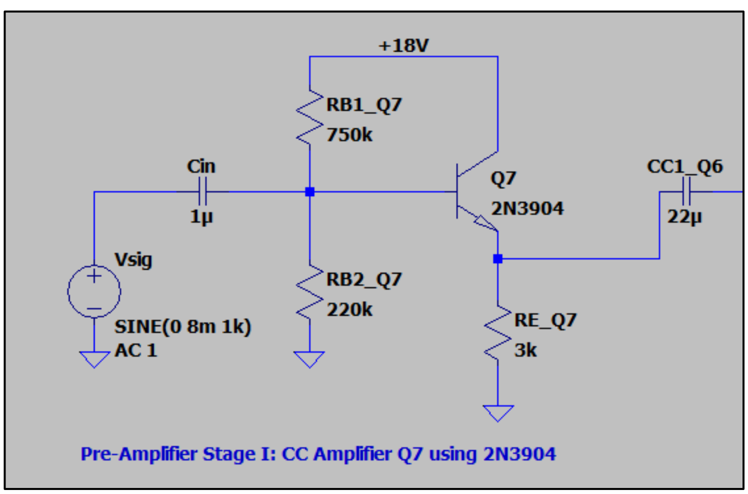
9. DRIVER / BIAS NETWORK

The second CE stage provides sufficient drive, so a separate driver stage is not required. A biasing network ensures correct operation of the Class AB output transistors and stabilizes the quiescent point.

10. DESIGN CALCULATIONS

Pre-Amplifier Stage I: CC Amplifier Q7 using 2N3904



Given

$$V_{CC} = 18\,V I_C = 1\,mA \beta = 80 R_L = 2\,k\Omega$$

Emitter Current I_E

For a CC stage:

$$\alpha = \frac{\beta}{\beta+1} = \frac{80}{81} I_E = \alpha I_C = \frac{80}{81}\times 1\,mA I_E \approx 0.988\,mA \approx 1\,mA$$

Bias Divider Voltage

Base voltage is set by the divider:

$$V_B = V_{CC}\frac{R_{B2}}{R_{B1}+R_{B2}} V_B = 18\times\frac{220k}{750k+220k} V_B = 18\times\frac{220}{970} V_B \approx 4.08\,V$$

Emitter Voltage V_E

$$V_E = V_B - V_{BE} V_E = 4.08 - 0.7 V_E \approx 3.38\,V$$

Emitter Resistor R_E

$$R_E = \frac{V_E}{I_E} R_E = \frac{3.38}{0.988\times 10^{-3}} R_E \approx 3.42\,k\Omega \approx 3\,k\Omega$$

(Standard value selected)

Base Current

$$I_B = \frac{I_C}{\beta} I_B = \frac{1\times 10^{-3}}{80} I_B = 12.5\,\mu A$$

Bias Divider Current Check

$$I_{div} = \frac{V_{CC}}{R_{B1}+R_{B2}} = \frac{18}{970k} I_{div} \approx 18.6\,\mu A I_{div} \approx 1.5\, I_B$$

This is acceptable for a **common collector stage** because strong emitter feedback makes the bias less sensitive to base current variations.

Small-Signal Emitter Resistance

$$r_e = \frac{25\,mV}{I_E} r_e = \frac{25\times 10^{-3}}{0.988\times 10^{-3}} r_e \approx 25.3\,\Omega$$

Input Resistance of CC Stage

$$r_\pi = \beta r_e = 80\times 25.3 r_\pi \approx 2.5\,k\Omega$$

Effective emitter resistance seen by the transistor:

$$R_E' = R_E \parallel R_L R_E' = 3k \| 2k R_E' = 1.2\,k\Omega$$

Base input resistance:

$$R_{in,base} = r_\pi + (\beta + 1) R_E' R_{in,base} = 2.02k + 81\times 1.2k R_{in,base} \approx 122\,k\Omega$$

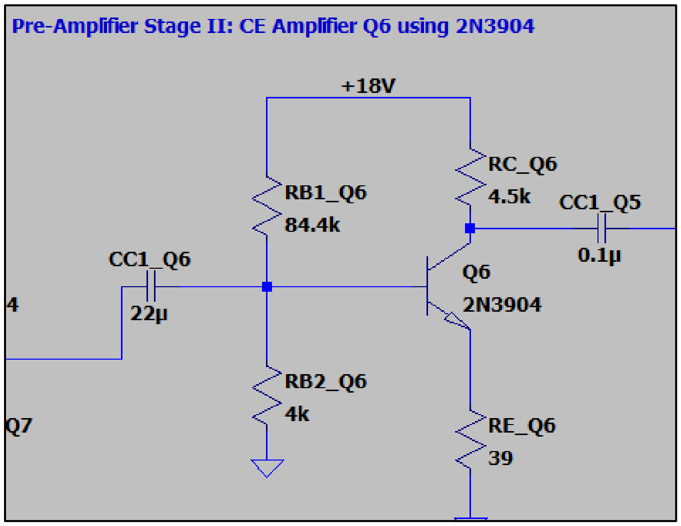
Total input impedance:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_{in,base} R_{in} = 750k\|220k\|122k R_{in} \approx 72\,k\Omega$$

Voltage Gain of CC Stage

$$A_v \approx \frac{R_E'}{R_E'+r_e} A_v = \frac{1200}{1200+25.3} A_v \approx 0.98 \approx 1$$

Pre-Amplifier Stage II: CE Amplifier Q6 using 2N3904



Given

$$V_{CC} = 18\,V I_C = 2\,mA \beta = 80 R_L = 1.6\,k\Omega$$

Target voltage gain:

$$A_v \approx 23\text{–}25\,V/V$$

Collector Voltage Selection

For maximum symmetrical output swing:

$$V_C = \frac{V_{CC}}{2} = 9\,V$$

Collector Resistor R_C

$$R_C = \frac{V_{CC}-V_C}{I_C} R_C = \frac{18-9}{2\times10^{-3}} R_C = 4.5\,k\Omega$$

Emitter Current I_E

$$\alpha = \frac{\beta}{\beta+1} = \frac{80}{81} I_E = \alpha I_C = \frac{80}{81}\times 2\,mA I_E \approx 1.98\,mA$$

Small-Signal Emitter Resistance r_e

$$r_e = \frac{25\,mV}{I_E} r_e = \frac{25\times10^{-3}}{1.98\times10^{-3}} r_e \approx 12.6\,\Omega$$

Effective Collector Load

$$R_C \parallel R_L = 4.5\,k \parallel 1.6\,k R_C \parallel R_L = \frac{(4.5)(1.6)}{4.5+1.6} R_C \parallel R_L \approx 1.18\,k\Omega$$

Emitter Resistance from Gain Requirement

Voltage gain of a CE stage with emitter degeneration:

$$A_v \approx \frac{R_C \parallel R_L}{r_e + R_E}$$

Rearranging to find R_E:

$$r_e + R_E = \frac{R_C \parallel R_L}{A_v}$$

Taking $A_v \approx 23$:

$$r_e + R_E = \frac{1180}{23} \approx 51.3\,\Omega R_E = 51.3 - 12.6 R_E \approx 38.7\,\Omega \approx 39\,\Omega$$

Emitter Voltage V_E

$$V_E = I_E \times R_E V_E = 1.98\times10^{-3}\times39 V_E \approx 0.077\,V$$

Base Voltage V_B

$$V_B = V_E + V_{BE} V_B = 0.077 + 0.7 V_B \approx 0.78\,V$$

Base Current

$$I_B = \frac{I_C}{\beta} I_B = \frac{2\times10^{-3}}{80} I_B = 25\,\mu A$$

Bias Divider Current Selection

Choose a stiff divider:

Taking:

$$I_{div} \approx 200\,\mu A$$

Bias Resistor Values

Lower resistor:

$$R_{B2} = \frac{V_B}{I_{div}} R_{B2} = \frac{0.78}{200\times10^{-6}} R_{B2} \approx 3.9\,k\Omega \approx 4\,k\Omega$$

Upper resistor:

$$R_{B1} = \frac{V_{CC}-V_B}{I_{div}} R_{B1} = \frac{18-0.78}{200\times10^{-6}} R_{B1} \approx 84.4\,k\Omega$$

Input Impedance of CE1 Stage

Small-signal base resistance of the transistor is given by:

$$r_\pi = \beta r_e r_\pi = 80\times12.6 r_\pi \approx 1008\,\Omega$$

The effective resistance looking into the base due to emitter degeneration is:

$$R_B = r_\pi + (\beta + 1) R_E R_B = 1008 + 81\times39 R_B = 1008 + 3159 R_B \approx 4167\,\Omega \approx 4.17\,k\Omega$$

The total input impedance of the CE1 stage is the parallel combination of the biasing resistors and the transistor input resistance:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_B R_{in} = 84.4k \| 4k \| 4.17k$$

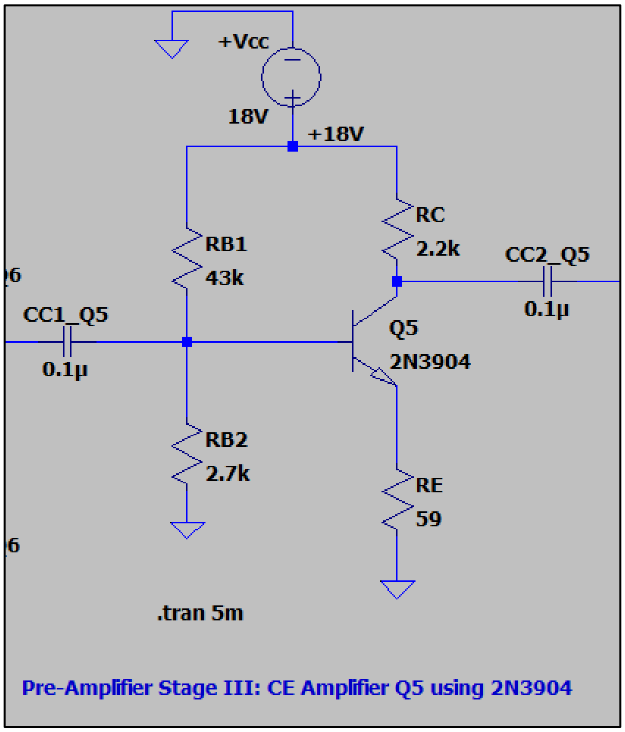
First combine R_{B2} and R_B:

$$4k \| 4.17k = \frac{(4)(4.17)}{4+4.17} = \frac{16.68}{8.17} \approx 2.04\,k\Omega$$

Now include R_{B1}:

$$R_{in} = 84.4k \| 2.04k R_{in} \approx 1.99\,k\Omega$$

Pre-Amplifier Stage III: CE Amplifier Q5 using 2N3904



Given

$$V_{CC} = 18\,V$$

$$I_C = 4\,mA \text{ (assumed)}$$

$$\beta = 80 \text{ (assumed)}$$

$$R_L = 5.4\,k\Omega$$

Collector Voltage Assumption

Assume the collector voltage at mid-supply for maximum symmetrical swing:

$$V_C = \frac{V_{CC}}{2} \approx 9\,V$$

Collector Resistor R_C

$$R_C = \frac{V_{CC}-V_C}{I_C} R_C = \frac{18-9}{4\times10^{-3}} R_C = 2.25\,k\Omega \approx 2.2\,k\Omega$$

(Selected standard value)

Emitter Current I_E

$$\alpha = \frac{\beta}{\beta+1} = \frac{80}{81} I_E = \alpha I_C = \frac{80}{81}\times 4\,mA = 3.95\,mA$$

Small-Signal Emitter Resistance r_e

$$r_e = \frac{25\,mV}{I_E} r_e = \frac{25\times10^{-3}}{3.95\times10^{-3}} r_e \approx 6.33\,\Omega$$

Effective Collector Load

$$R_C \parallel R_L = 2.2\,k \| 5.4\,k R_C \parallel R_L = \frac{(2.2)(5.4)}{2.2+5.4} R_C \parallel R_L \approx 1.56\,k\Omega$$

Emitter Resistance from Gain Requirement

Voltage gain of a CE stage with emitter degeneration is:

$$A_v \approx \frac{R_C \| R_L}{r_e + R_E}$$

Rearranging to find R_E:

$$r_e + R_E = \frac{R_C \| R_L}{A_v} r_e + R_E = \frac{1560}{24} = 65\,\Omega R_E = 65 - 6.33 R_E \approx 58.7\,\Omega \approx 59\,\Omega$$

Emitter Voltage V_E

$$V_E = I_E \times R_E V_E = 3.95\times10^{-3}\times59 V_E \approx 0.233\,V$$

Base Voltage V_B

$$V_B = V_E + V_{BE} V_B = 0.233 + 0.7 V_B \approx 0.93\,V$$

Base Current

$$I_B = \frac{I_C}{\beta} I_B = \frac{4\times10^{-3}}{80} I_B = 50\,\mu A$$

Bias Divider Current

Choose:

$$I_{div} = 10 I_B = 500\,\mu A$$

Bias Resistor Values

Lower resistor:

$$R_{B2} = \frac{V_B}{I_{div}} R_{B2} = \frac{0.93}{500\times10^{-6}} R_{B2} \approx 1.86\,k\Omega$$

Upper resistor:

$$R_{B1} = \frac{V_{CC} - V_B}{I_{div}} R_{B1} = \frac{18-0.93}{500\times10^{-6}} R_{B1} \approx 34.1\,k\Omega$$

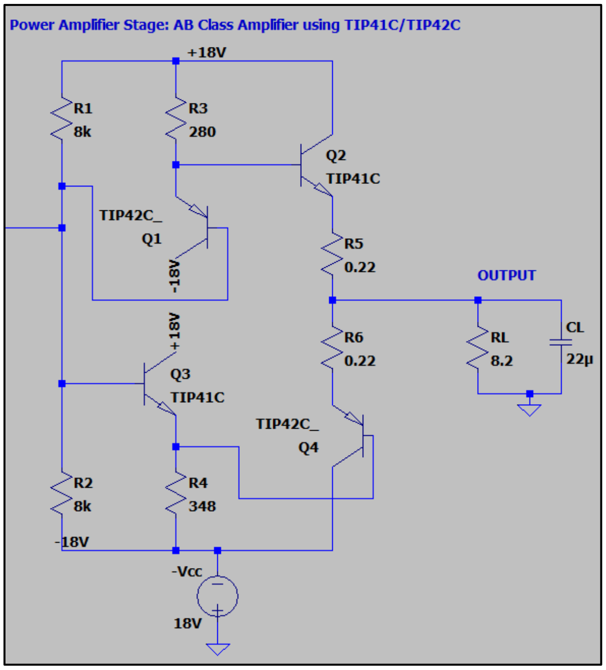
Input Resistance of CE Stage

$$r_\pi = \beta r_e = 80\times6.33 = 506\,\Omega R_B = r_\pi + (\beta + 1) R_E R_B = 506 + 81\times59 R_B \approx 5.29\,k\Omega$$

Total input resistance:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_B R_{in} = 34.1k \| 1.86k \| 5.29k R_{in} \approx 1.6\,k\Omega$$

Power Amplifier Stage: AB Class Amplifier using TIP41C/TIP42C



Assuming $\beta = 30$ for TIP 41C, TIP 42C

Assuming $V_{EB} = 0.6V$

$$i_L = \frac{5}{8.2} = 0.6098\ A \approx 0.61\ A$$

So As only one of Q_2 and Q_4 are on at a time so they must supply 0.61 A during their on cycle.

$$ie_{Q2} = 0.61A$$

$$ib_{Q2} = \frac{0.61}{30} A$$

$$= 19.7mA$$

By rule of thumb for Q_1 to supply 19.7mA,

$$ic_{Q1} = 2.5\ x\ ib_{Q2}$$

$$= 49.25mA \approx 50mA$$

$$ib_{Q1} = \frac{50m}{30} A$$

$$= 1.667mA$$

$$R_3 = \frac{18 - V_{EB}}{50m} = 348\Omega$$

Similarly for Q_2 and Q_4

$$ie_{Q4} = 0.61A$$

$$ib_{Q2} = 19.7mA$$

$$ic_{Q2} = 2.5\,x\,ib_{Q4} = 49.25mA \approx 50mA$$

$$R_4 = \frac{-0.6-(-18)}{50mA} = 348\Omega$$

$$ib_{Q2} = \frac{50m}{30}A$$

$$= 1.667mA$$

$$R_1 = R_2 = \frac{18}{1.667m} = 10.797k\Omega$$

R_5 and R_6 are chosen to reduce effects of V_{BE} and V_{EB} mismatches and thermal runaway and taken as 0.22Ω each based on standards.

$$R_5 = R_6 = 0.22\Omega$$

$$R_{in} = 10.8k \| R_B$$

$$R_B = r_\pi + (\beta + 1)R_E$$

$$R_B = (\beta + 1)r_e + (\beta + 1)R_E$$

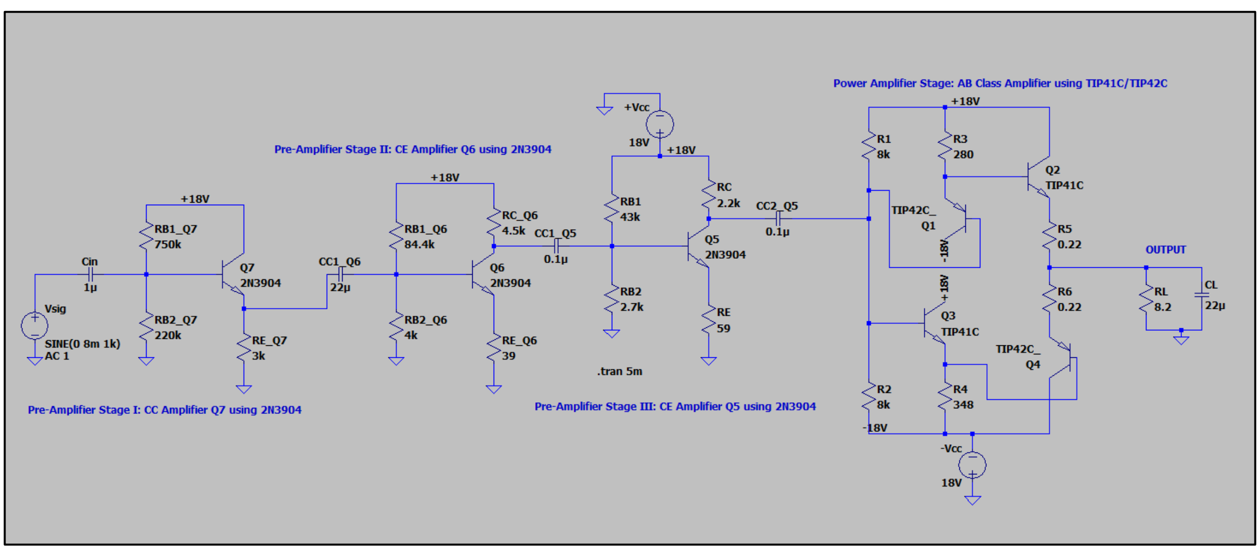
$$re \approx \frac{25\,mV}{I_E} = \frac{25\,mV}{48.39mA} = 0.517$$

$$R_B = (31\,x\,0.517) + (\,31\,x\,348) = 10.805k\Omega$$

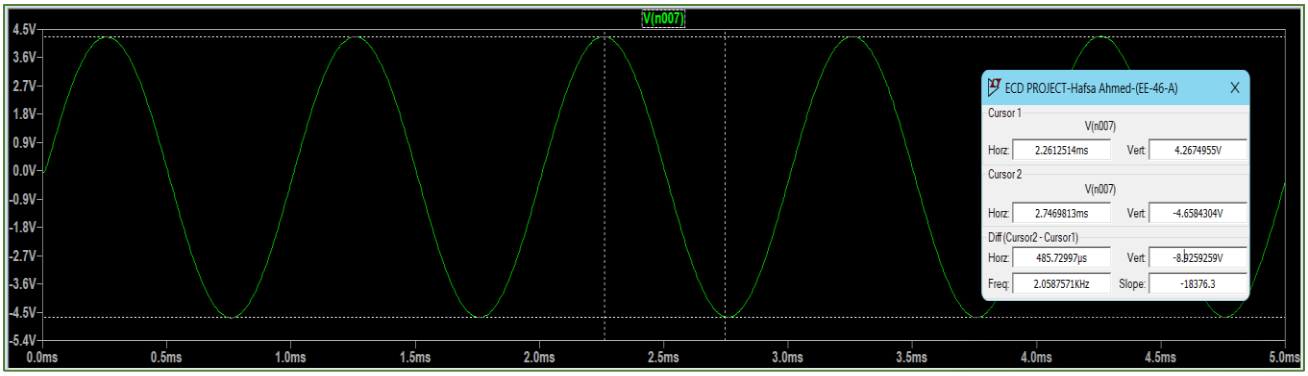
$$R_{in} = 10.8k \| (10.805k) = 5.4k\Omega$$

$$R_{in} = R_{L\,of\,prev\,stage} = 5.4k\Omega$$

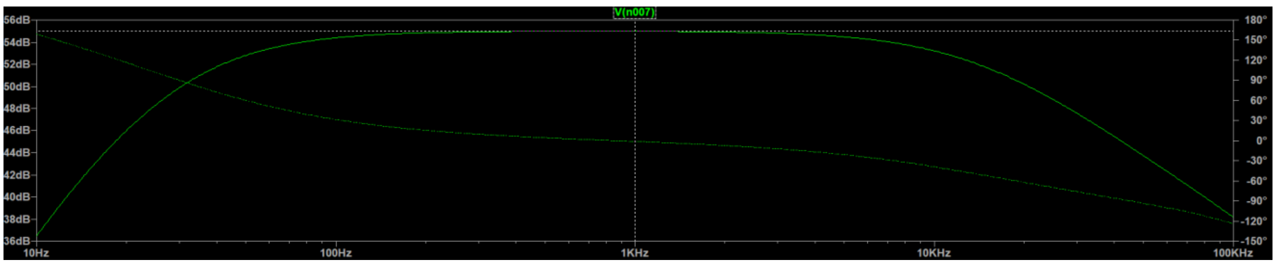
LtSpice-SIMULATIONS



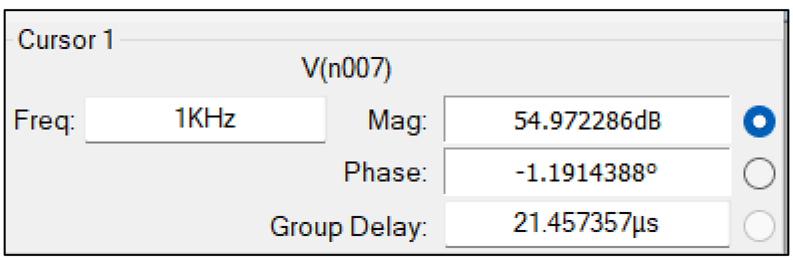
OUTPUT WAVEFORM



BODE PLOT

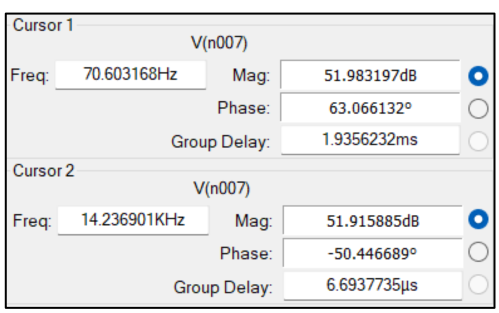


Midband Gain in dB



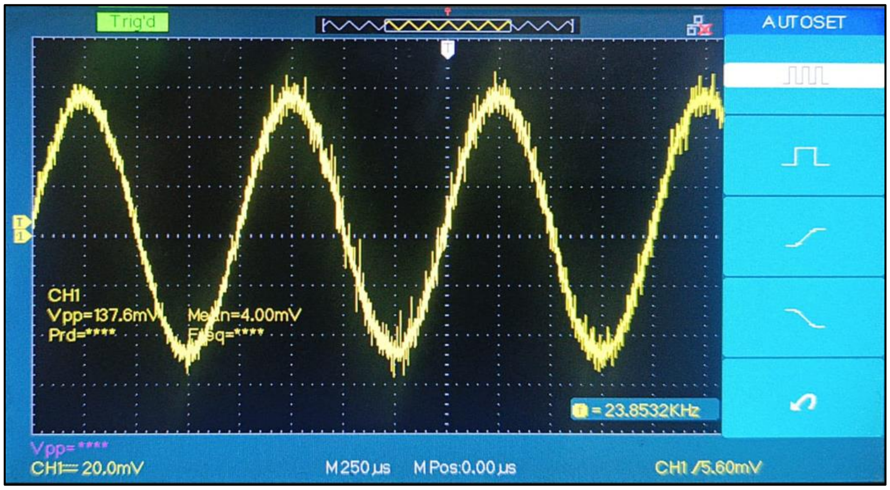
F_L and F_H – 3dB

$$54.972 - 3 = 51.972dB$$

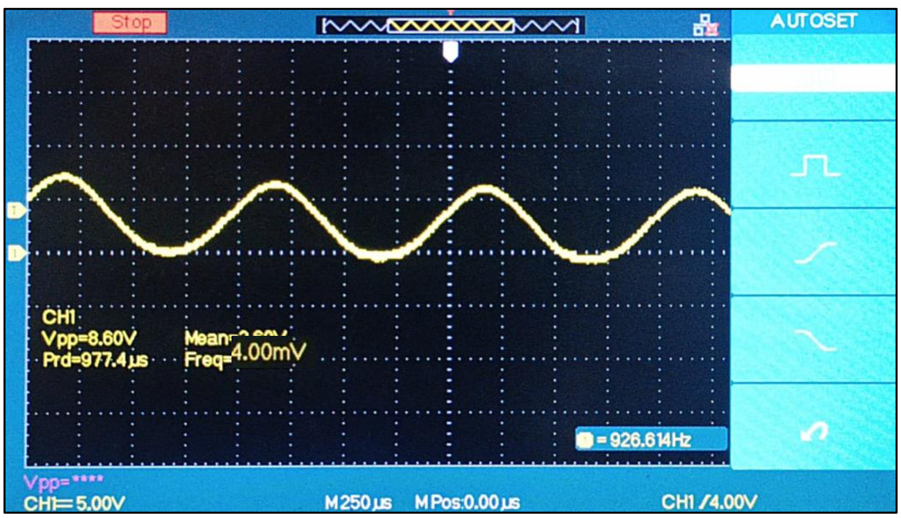


OSCILLOSCOPE SIGNALS/HARWARE IMPLEMENTATION

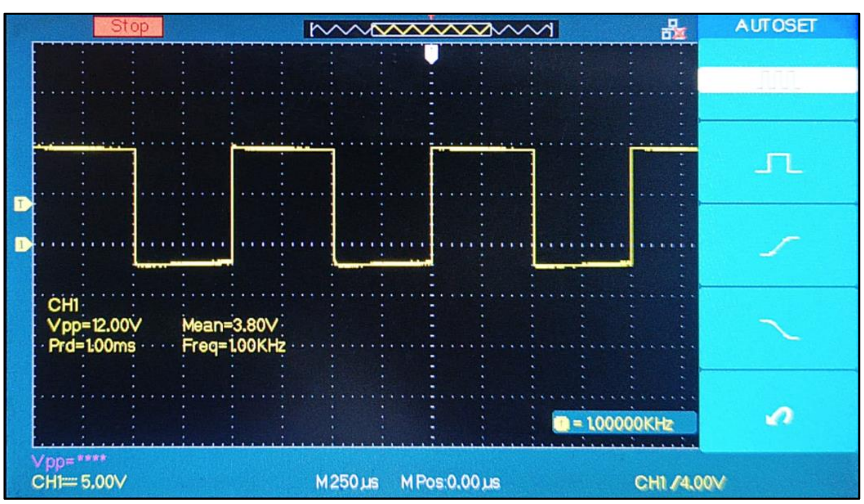
Input Signal

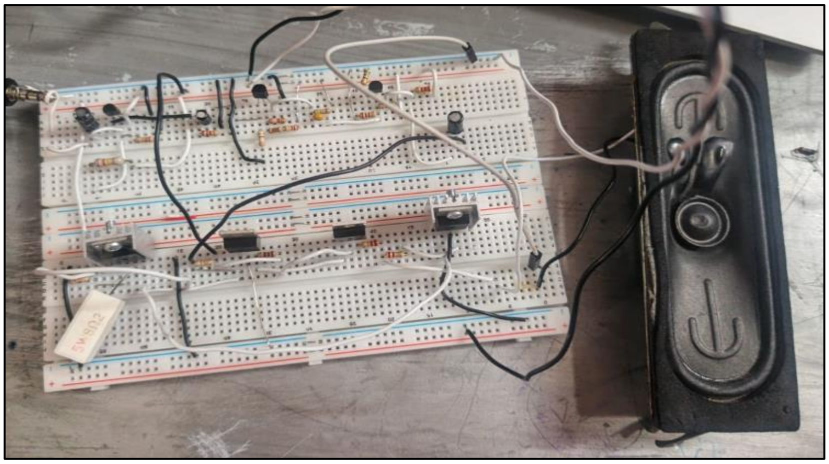


Output Signals – Av TEST



B) Frequency Response Test





Conclusion:

Designed discrete BJT audio amplifier successfully achieves a voltage gain greater than 500 V/V while maintaining stable operation over the 100 Hz to 12 kHz audio range. The Class AB push-pull output stage effectively drives an 8 Ω speaker with low distortion and good efficiency. Overall, both simulation and practical results confirm that the amplifier meets the required specifications using a fully discrete transistor-based approach.